NEUMA · HUMAN PRACTICES

DIAGNOSIS CULTURE IN MEXICO (PUEBLA)

Interpretive paradigm · Qualitative approach

25 clinical specialists

2 public hospitals

3 focus groups

20 voices

8 biosensor experts

Hospitals IMSS HGZ No. 20 “La Margarita” · ISSSTE Puebla
Experts INAOE · UNAM · industry



CONTEXT

77.6% of the population is publicly insured
Commonwealth Fund, 2026

82% of NSCLC cases are diagnosed at a metastatic stage
Añorve Bailon et al., 2024

> *“Culture is an object of interpretation”*
> Our lens, after Geertz (1973)

Human Practices · wiki

Interview / focus group photo
only with image consent

Interview / focus group photo
only with image consent

Interview / focus group photo
only with image consent

Map: svg-maps/mexico (CC BY 4.0), schematic inset

WHAT WE HEARD

Late diagnosis is systemic, not a lack of education

THE SYSTEMIC ICEBERG



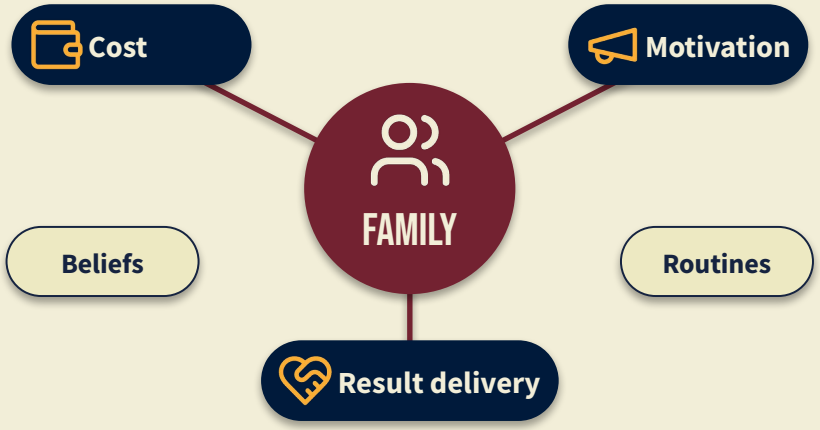
THE PATIENT PATHWAY (PUBLIC INSURANCE)



IN THEIR WORDS AND OURS

"Just a little cough"	The patient is the family's provider
3 to 8 months to a diagnosis	No national lung cancer screening program
"It happens to smokers"	*"Diagnosis" sounds like a "verdict"*
People act when someone they trust tells them to	Family pays · pushes · receives the result

CULTURE MAP · SIMPLIFIED



Diagnosis culture: shared beliefs, expectations and routines

> *"I feel sick, but I'll go tomorrow because I have to work; otherwise, what will my family eat?"*
> Jorge Jiménez Vargas · Nurse, HGZ No. 20

> *"A strategy as such? No… it is almost always a finding rather than an early detection."*
> Dr. Gilberto Pérez Luna · Epidemiology, ISSSTE

> *"Now everyone is prone."*
> Focus group 3 · working adults, Puebla

NEUMA · HUMAN PRACTICES

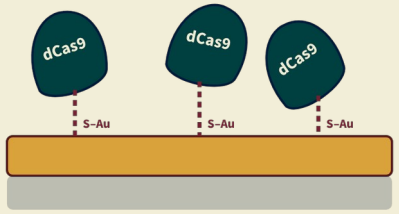
HP REACHED THE BENCH

Every requirement has a name

INAOE · JUNE–JULY 2026

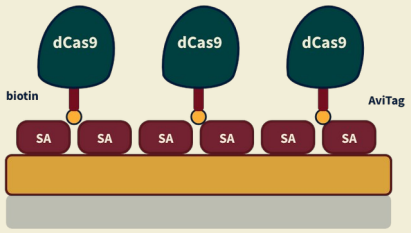
Immobilization redesigned — Cys–S–Au → Biotin–SA

BEFORE · CYSTEINE ANCHORING



Gold surface · random orientation

AFTER · AVITAG–BIOTIN / STREPTAVIDIN



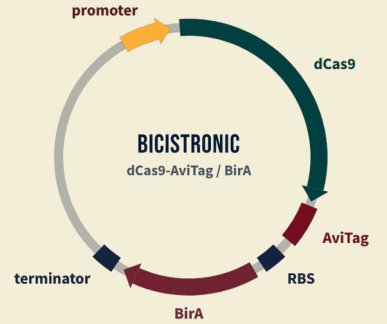
Streptavidin SPR chip · oriented dCas9

INAOE Cysteine anchoring on gold → AviTag-biotin and streptavidin · validated streptavidin SPR chips

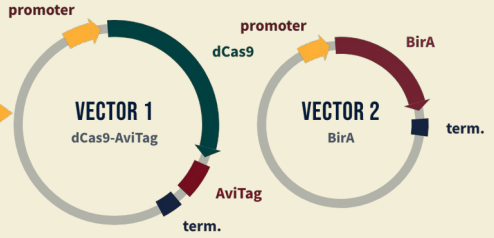
UNAM · SEPTEMBER 2026

Expression redesigned — 1 vector → 2 vectors

BEFORE · ONE BICISTRONIC VECTOR



AFTER · TWO SEPARATE VECTORS



Schematic · not to scale · backbone elements omitted

UNAM Bicistronic dCas9-AviTag/BirA → two separate vectors · in vitro biotinylation

HP IN LAB · REQUIREMENT → TECHNIQUE → WHO RAISED IT

Requirement	Lab technique	Raised by
Reliable	Three controls · several gRNAs screened	Dr. Irma Cruz Solís (controls) · biosensor workshop experts (gRNAs)
Specific	dCas9 recognition checked per target	Biosensor workshop experts
Reproducible	Standardized expression and purification	Biosensor workshop experts · Dr. Armando Hernández García, UNAM
Affordable	Simpler, cheaper SPR chip	Workshop experts (6 of 8) · nursing and internal medicine, HGZ No. 20
Low burden	Single blood sample · ctDNA liquid biopsy	Dr. Gilberto Pérez Luna, ISSSTE · focus group 2